



Operational Concept Description
for the
KNEAD Example System

DCN: KNEADOC20240103-P1:128

Revision Date: 14 Mar 2024

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Controlled by: AGENCY
Controlled by: OFFICE
CUI Category: WILL-BE
Limited Dissemination Control: TBD
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Distribution Statement D: Vinay Agarwal



DOCUMENT CHANGE HISTORY

The following table is a simple list of released revisions sent for review. Records of reviews and the review artifacts are saved with reviewer information in the The Balance Project artifact repository.

Change Record

Date	Version	Author(s)	Change Reference
13 Sep 2025	P1	Vinay Agarwal	Preliminary DRAFT version
5 Dec 2025	P1	Vinay Agarwal	Final Draft of the system

Draft P1 Preliminary version of this document.

1. Baseline Document draft 1
2. Final fix and update from the first draft



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CHAPTER 1

Scope

This document provides the Operational Concept Description (**OCD**) for the Balance System. The system will be referred to as the Balance System.

1.1 Identification

The Balance System described in this document shall be known as Balance System version 1. However, the Operational Concept Description **OCD** described herein shall be applicable to pre-releases such as Beta-releases for a phased release as listed for each requirement. The major system interfaces and capabilities are fully specified in Chapter 3.

1.2 System Overview

Figure 1 shows the high-level architecture for the Balance System system. This diagram shows the major external interfaces that provide the capabilities of Balance System.

The Balance System is a single player game. The game is a standalone game played on the STM32 board. This system would be a game where the user would have to balance a ball on a LCD screen that is builtin on the STM32 board. The objective of the game is to balance the ball on the screen based on the way the board was tilted. Using an onboard gyroscope that communicates across the board to the MCU though SPI communication. Balance System would keep track of the current position of the ball and where the next updated move is. This helps keep track of the system of where the ball is until a movement has occurred. Balance System shall process at a maximum 180 Hz. This would give the user enough time to process the current angle of the ball and be able to present on the LCD-TFT screen.

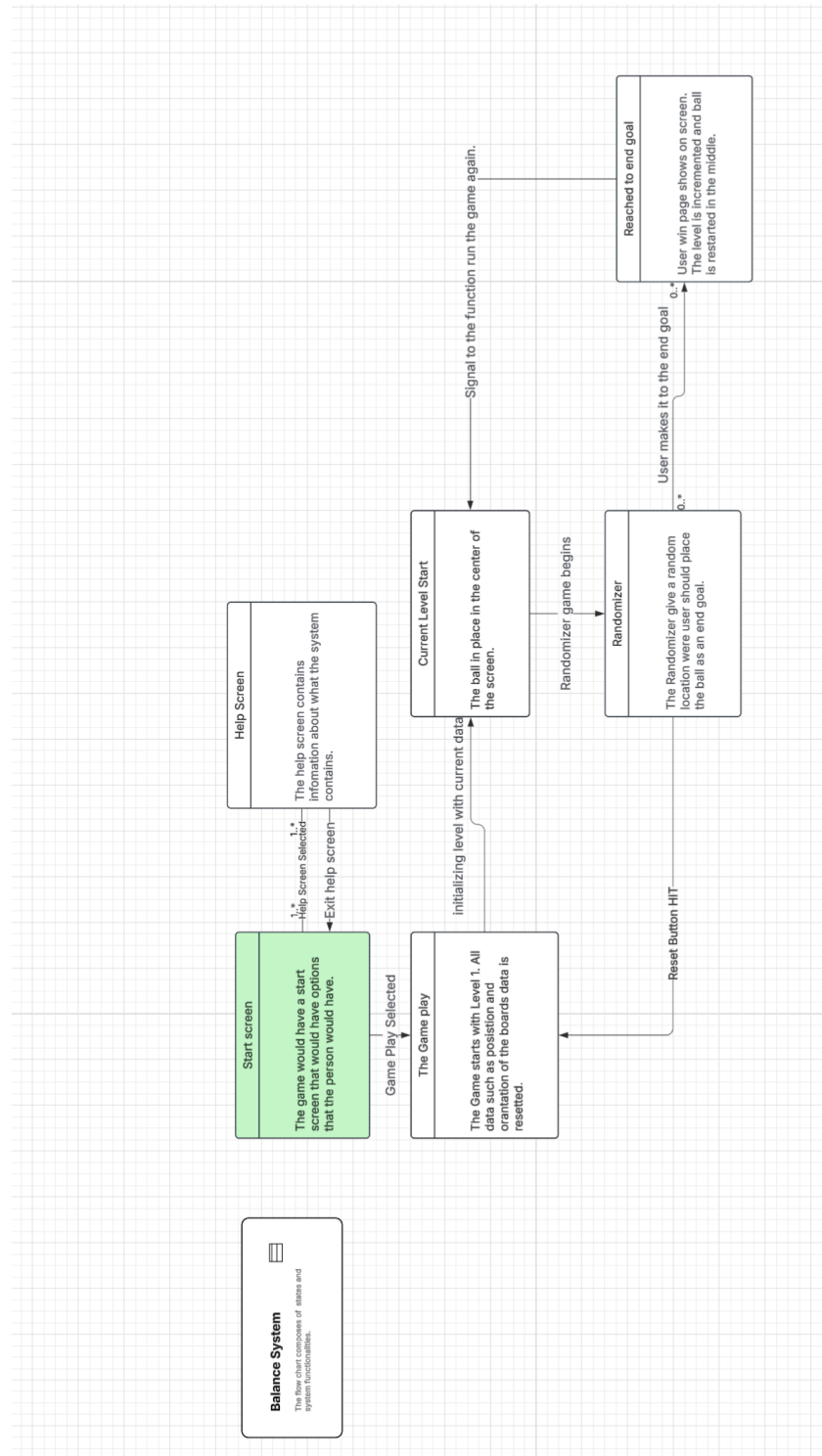


Figure 1: System Overview



1.3 Document Overview

This section provides information about this document's security/privacy considerations, contents, structure, and version information.



CHAPTER 2

References

This section provides a list of referenced items for this document.

2.1 Acronyms and Abbreviations

This section defines acronyms and abbreviations used in this and related documents.

Table 1: Acronym Definitions

Acronym	Definition
CONOP	Concept of Operations
DID	Data Item Description
OCD	Operational Concept Description
POC	Point of Contact
SPS	System Performance Specification
SSS	System / Subsystem Specification
End of acronym definition table	

2.2 Glossary and Definitions

This section defines glossary terms used in this and related documents.

Table 2: Glossary Terms and Definitions

Glossary Term	Definition
STM32F429I	Micro-controller board has all component fit onto one board.
Customer	The professor that is view the grading all assignments.
End of glossary terms table	



2.3 Referenced Documents

This section lists the referenced documents for this document. The references are categorized into two categories:

External Documents not directly associated with this project.

Project Documents that are directly associated with this project.

2.3.1 External Documents

2.3.2 Project Specific Documents



CHAPTER 3

Current system or situation

3.1 Background, objectives, and scope

Balance System is a user friendly game. The objective is for the user to move a ball from an initial position to a new position. Each Level the goal position will move to a different location. Users who play this game will have a different experience then an iphone app. This application will only be used will out other systems running on the platform.

3.2 Operational policies and constraints

The limitations of the system are the speed of sampling from the MEMs chip and speed of displaying position of the ball. The physical system would be connected to an usb cable that would provide the system with power.

3.3 Description of current system or situation

Moving the ball would require the user to tilt the board at a certain angle. This angle should be calculated based on the MEM existing on the board. Once the board goes through a startup states the system shall show main menu. If the users wants to improve there understanding of how to game works, there is a help tab. If the user would like to start the game, they can click the "Start Game". Once the user starts the game the user would increase every level sequentially. If the user wants to reset the game, they can select the "Reset Game" option from the main menu. This will return the ball to its initial position and reset all game levels and scores to their default state. The system will prompt the user for confirmation before performing the reset to prevent accidental loss of progress.

3.4 Users or involved personnel

Currently, there is only one user that is involved with this project. However, the developer has been using open source tools.

3.5 Support concept

The support for this system should be set as a minimal. Maintenance team shall look for release updates and follow installation instructions with the procedure documentations.



CHAPTER 4

Justification for and nature of changes

The future updates to the game have the ability to update the change the motion of the "BALL" displayed.

4.1 Justification for change

Balance System would be and upgrade from normal games that user play on there phone.

4.2 Description of needed changes

Upgrading the system would allow the user have mor flexibility to move the system.

4.3 Priorities among the changes

DIDINFOOCD-4.3 :: This paragraph shall identify priorities among the needed changes. It shall, for example, identify each change as essential, desirable, or optional, and prioritize the desirable and optional changes.

This allows the developer to have more of a better understanding of the system.

4.4 Changes considered but not included

The changes allow the user only play the game that is provide on the board. Another addition to the system could be to add a Quadiped and create a closed looped system.

4.5 Assumptions and constraints

The SDRAM that the boards memory has.



CHAPTER 5

Concept for a new or modified system

This chapter talks about additional improvements to the current system. What I would like to add on top of this project is to have a quadratic robot move in such a direction to stabilize the ball on the screen. That current system is not been implemented yet, and therefore we are only using the gamer experience.

5.1 Background, objectives, and scope

The original objective of this project was to have a quadratic robot balance a ball on an LCD screen through the STM 32 board. As the semester went on, I realize that was more optimistic than I could manage. Therefore, I just stuck to making it a game.

5.2 Operational policies and constraints

The operational policies in the constraints two those would be that the user would have to be clear about how the quadratic would've worked and also how it incorporates into the system because there is a lot of issues that may occur. These issues would be that the motors wouldn't spin properly or they would be a lot of weird movements that would happen or issues with having the robot listen to commands.

5.3 Description of the new or modified system

These new modifications would have not been implemented within the time. We would want, however we still have created space to add additional modifications to the system.

5.4 Users/affected personnel

My personal effects to the system would have been my own microcontroller being able to communicate through all the motors and how they all would work together to take commands from the system and make a decision. I would've also added a type of inference model to determine the best outcome movement for that system.

5.5 Support concept

This project doesn't have any support concepts for the additional portion of the project.



CHAPTER 6

Operational scenarios

The operation of the new system will be used to drive game movement and foundational concepts to the gaming industry. Because the system will rely on acceleration for the ball movement, it would be a completely different process that the system will have to adapt to in order for the user to play the game properly.

6.1 Use Case 1

This System would allow to understand the make of the interactive gaming industry.

6.2 Use Case 2

To be able to do an AI model with a scream update of a game.



CHAPTER 7

Summary of impacts

The impacts that the system will have on the user will differ from common gaming system. This common gaming system allows. The user user to use a standard control controller. With with this current operation of the system, we allowed to use it to only have two loads of operation, which is clicking the screen or tilting the screen. This allows the use it to have more hands-on experience with the game. If we also added the additional feature of a robotic quadruped, then we could have the use of move them robot with a controller, but they would be controlling the robot and not just the game which allows them more of a physical movement to win the operation.

7.1 Operational impacts

These operational impacts on the user will allow them to interact with the game more clearly. This also allows the user to be more accurate when trying to move the ball in the right direction. If added more implementation was added into the system, then we can have it where the user would interact with the robot to be able to move the ball in the center.

7.2 Organizational impacts

There is no organizational impact that are going to affect the user.

7.3 Impacts during development

The biggest impact during the development stage of the product would be how the UI was formatted and it's interaction with the gyroscope. The reason why this would be a big issue because of speed and and representation on the screen how much do you tilt the screen and how much do we wanna see the ball change directions?



CHAPTER 8

Analysis of the proposed system

8.1 Summary of advantages

The new system shall have advance capabilities that other gamings systems don't contain. This system allow the user to engage with the game and enjoy each level.

8.2 Summary of disadvantages/limitations

The disadvantages of the system that the board game must be plugged to a power source in-order to play. Another issues is that this system is created as an NRE. The NRE is show as one of one. Therefore this board is the only board that change be used to play this game.

8.3 Alternatives and trade-offs considered

The trade offs between using this board and playing this game it the usage of that board it self. The test board has other options for usage. However, everything the game is needed is done with external peripheral.



APPENDIX

Notes

This section provides notes, as necessary, to document the system segmentation specification.



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